



THE UNIVERSITY OF ARIZONA

Responsible Artificial
Intelligence

Wolfgang Jentner

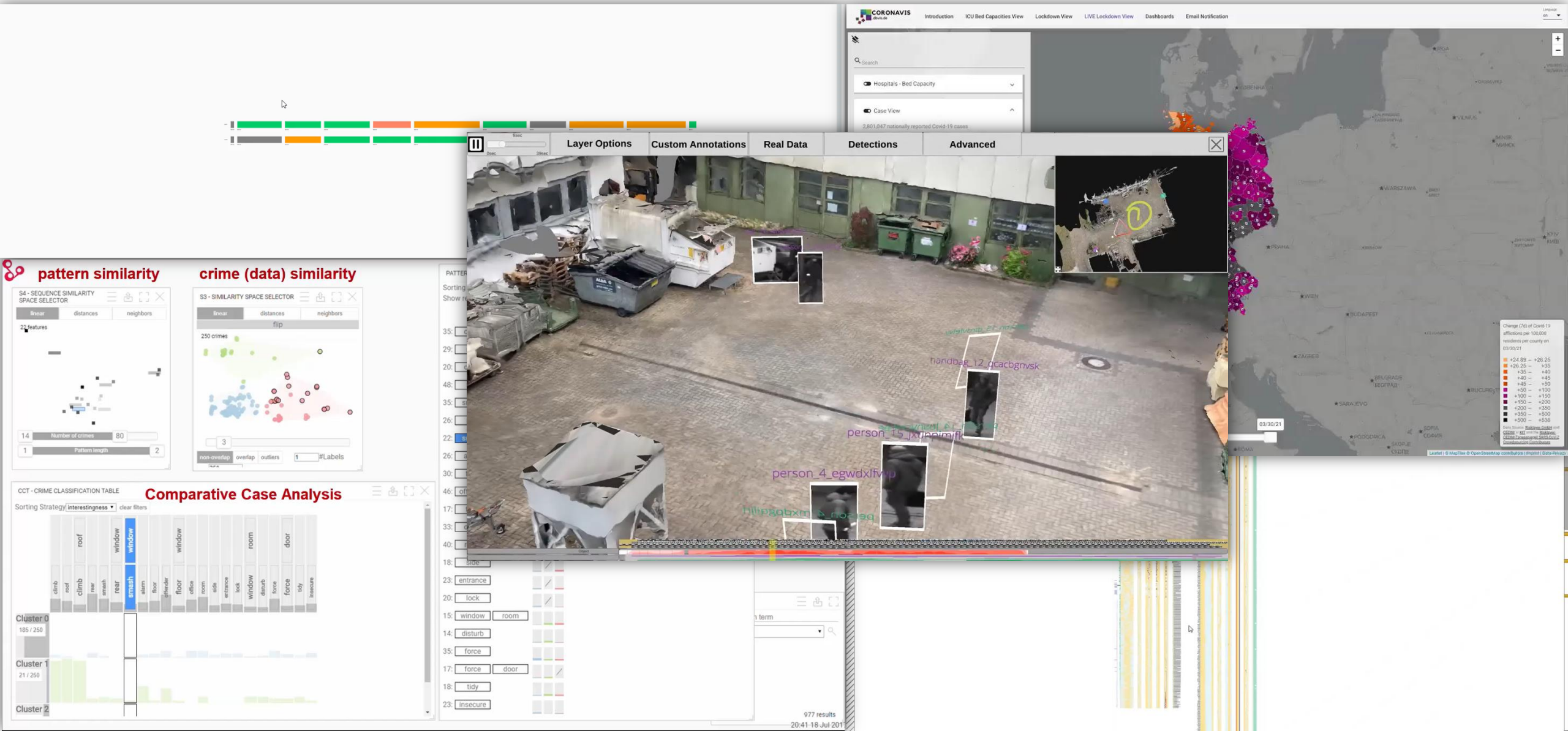
Arizona Institute for
AI and Society: AI²S

Responsible AI: Transparency, Accountability, Bias

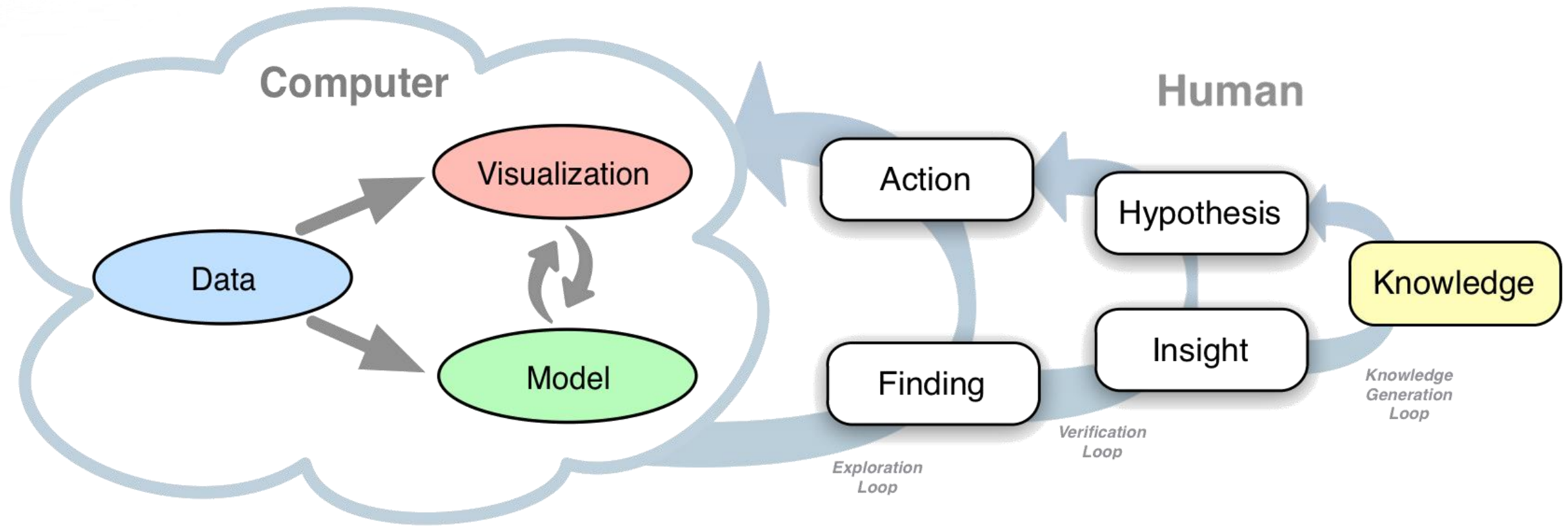
June 10th 2026



Arizona Institute for AI and Society: AI²S



Knowledge Generation Model for Visual Analytics

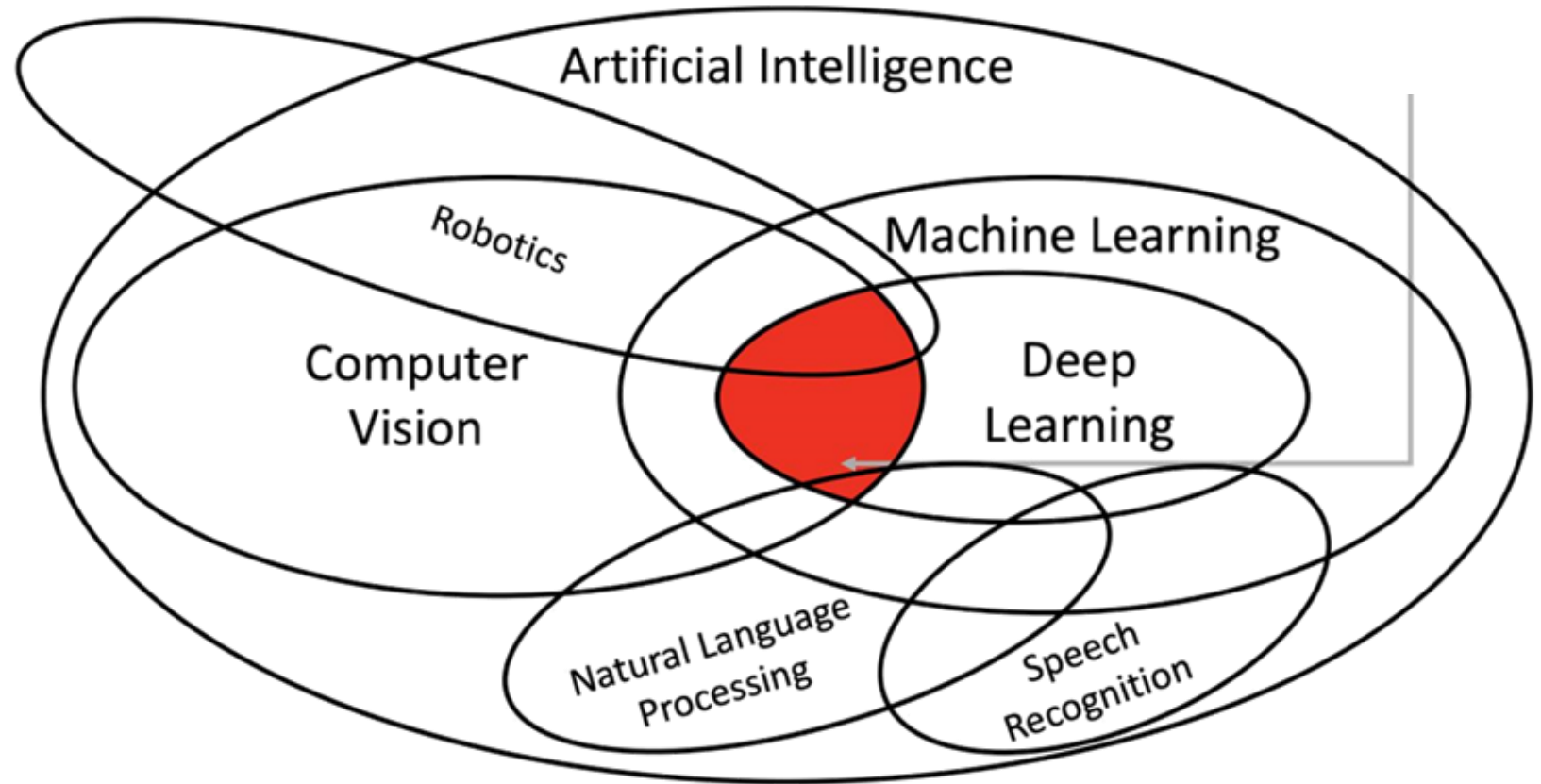


Artificial Intelligence

What is „AI“?

Automation of
intelligent behavior

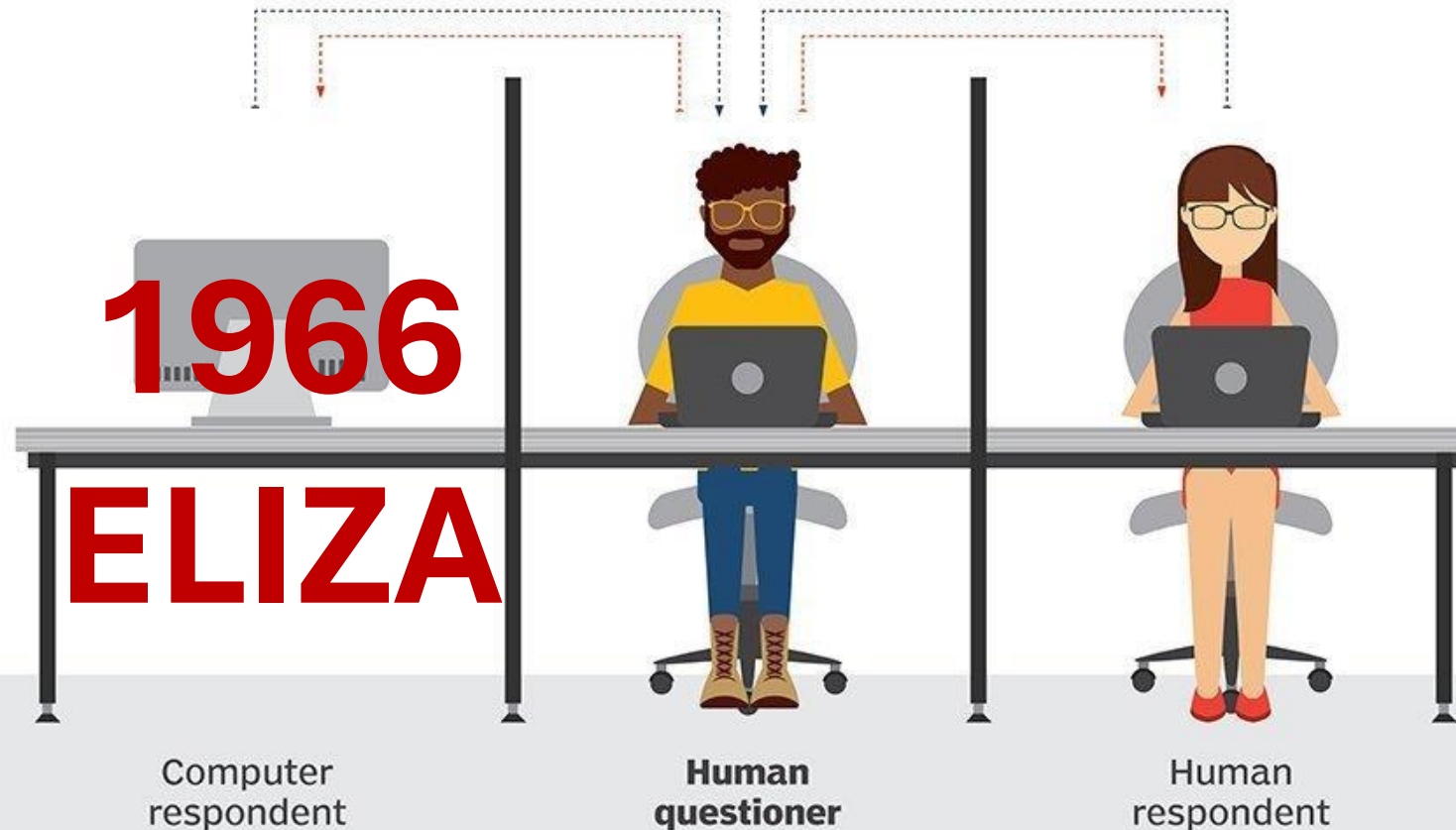
➔ no clear definition
for intelligence



Turing test 1949

During the Turing test, the human questioner asks a series of questions to both respondents. After the specified time, the questioner tries to decide which terminal is operated by the human respondent and which terminal is operated by the computer.

■ QUESTION TO RESPONDENTS ■ ANSWERS TO QUESTIONER



Intelligence?

Traits:

- Learning
 - Generalization
- Reasoning
 - Deductive
 - Inductive
- Problem solving
- Perception
- Using language



The Turing Test
1950

The term "artificial intelligence" is coined
1956

The term "machine learning" is coined
1959

The term "AI winter" is coined
1984

CNNs applicability
demonstrated
1989

Deep Blue defeats
Garry Kasparov
1997

A Neural Probabilistic
Language Model
2000

Turing Natural Language Generation
2019

Apple releases Siri
2011

ChatGPT
2022

1950

1957

1964

1971

1978

1985

1992

1999

2006

2013

2020

SNARC
1951

ELIZA and Shakey
1966

Samuel Checkers-Playing Program
1952

Artificial Intelligence: A General Survey
1973

The Logic
Theorist
1956

The General Problem Solver
1957

The first AI winter begins
1974

Symbolics Lisp machines commercialized
1980

Fifth Generation Computer Systems
1982

Large-Scale Deep Unsupervised Learning
2009

Bayesian networks causal analysis
1985

A Statistical Approach to Language Translation
1988

Watson wins on Jeopardy!
2011

Googles's deep neural network
project
2012

China's Tianhe-2 fastest system
2013

Facebook introduces DeepFace
2014

AlphaGo wins at Go
2016

Sophia: the first robot
with citizenship
2017

Google introduces the Transformer
2017

Cimon, GPT, Lovot
2018

Curial GPT-3 LLM
2020

Dall-E
2021

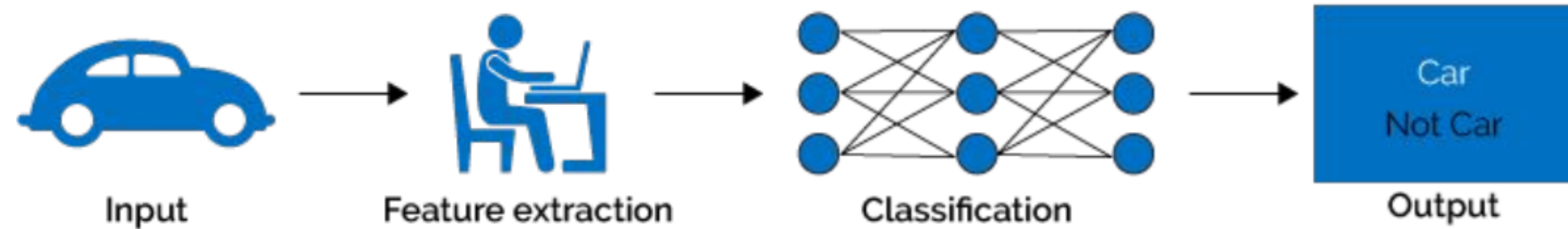
GPT-4
2023

History of Artificial Intelligence (AI) timeline

Source: <https://www.lucensoftware.com/blog/artificial-intelligence-ai-and-chatgpt-history-and-timelines>

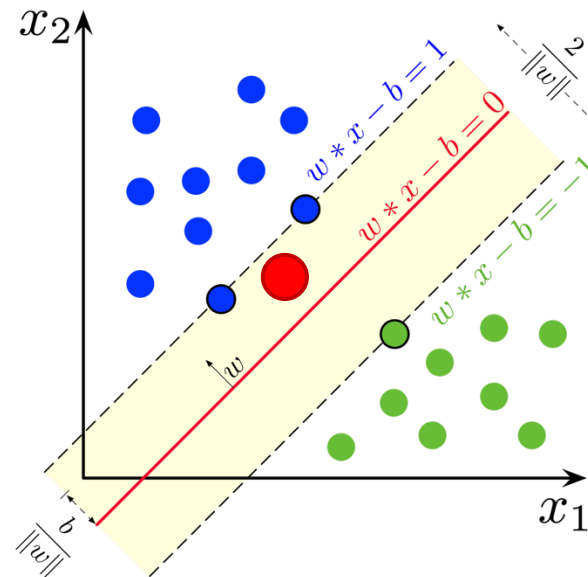
Machine Learning, Deep Learning & Transparency

Machine Learning



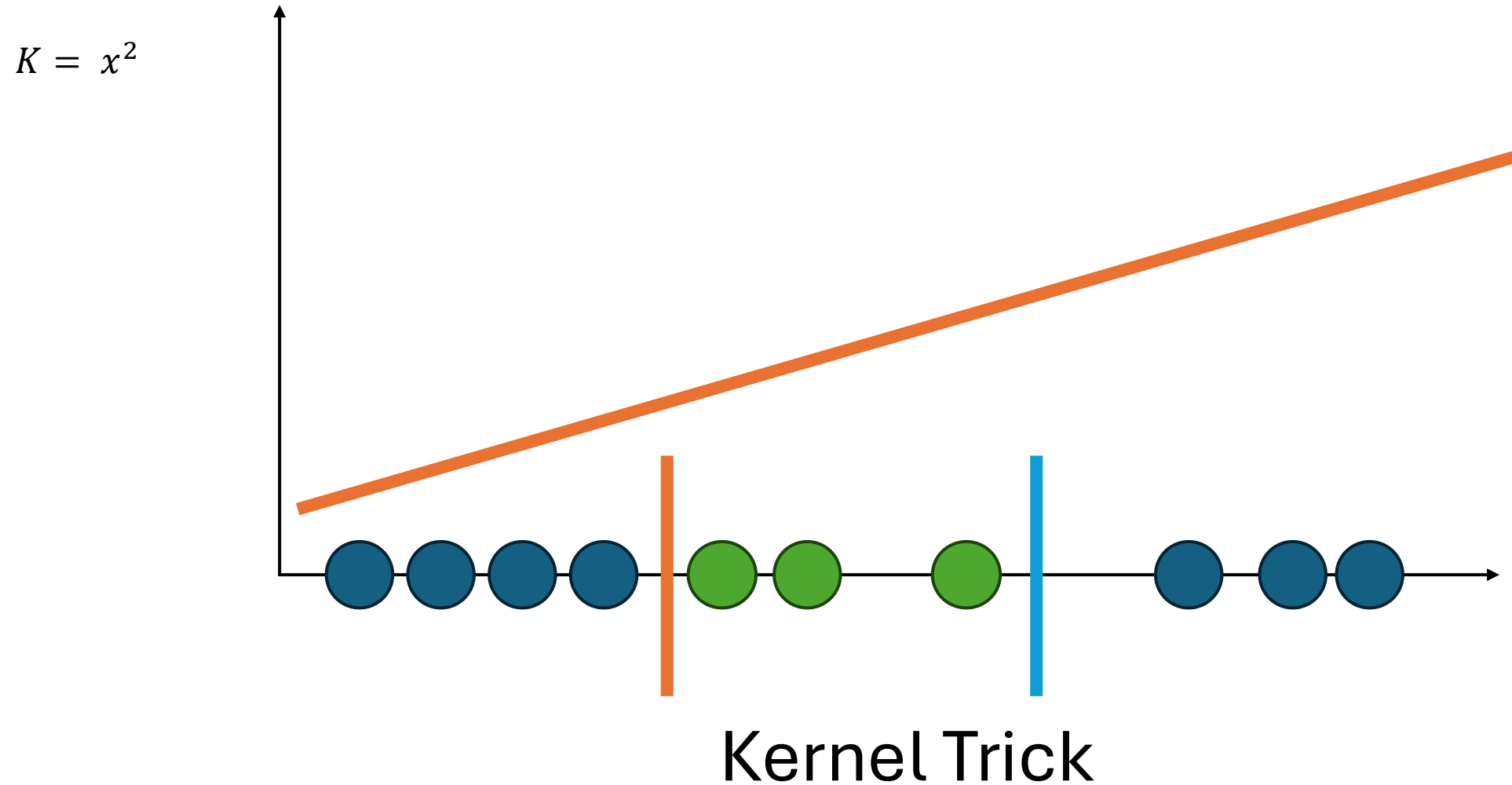
ML Example: Support Vector Machine

Features			Labels
# Doors	# Wheels	Engine?	Car?
5	4	1	Yes
3	4	1	Yes
0	2	1	No
2	3	1	?

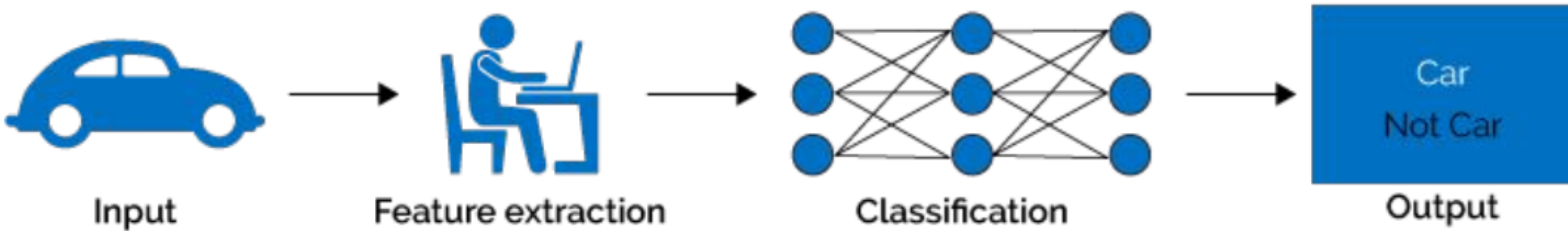


Prediction: Car

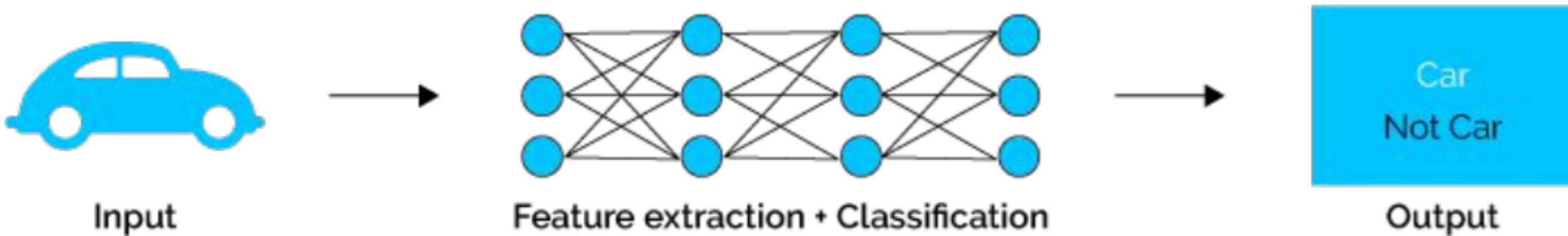
ML Example: Support Vector Machine



Machine Learning

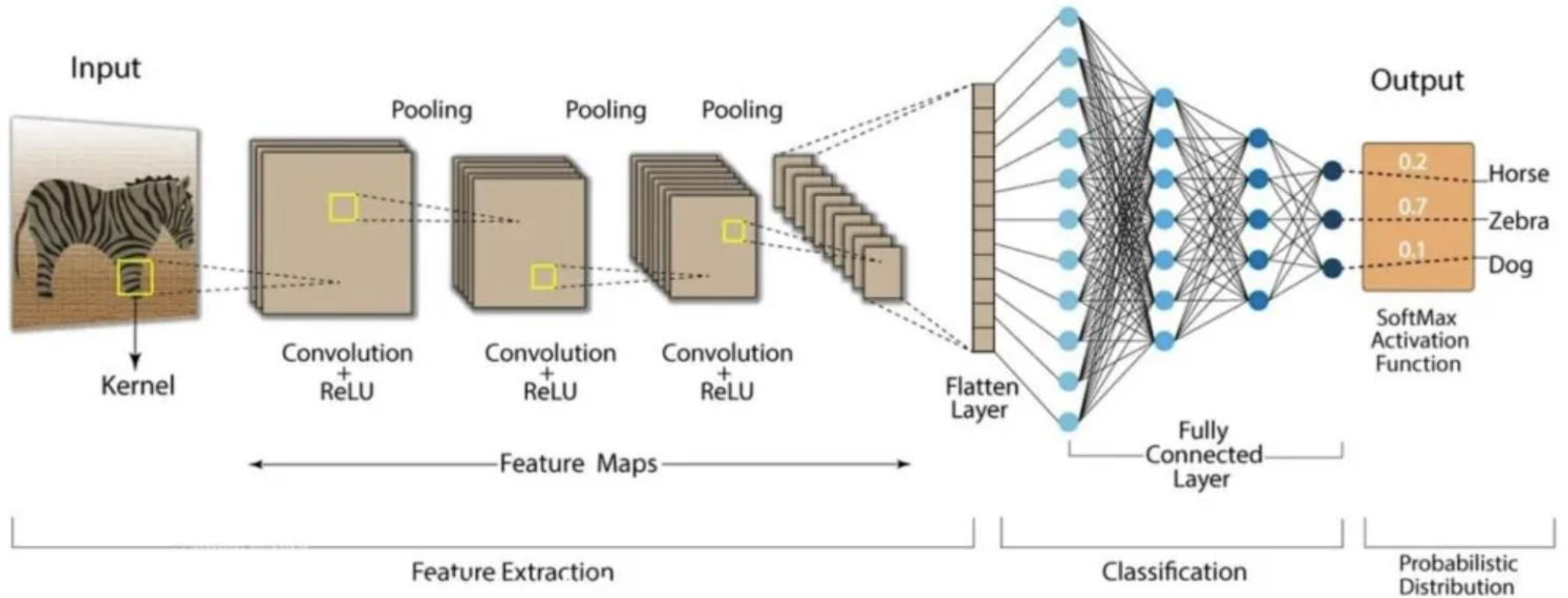


Deep Learning

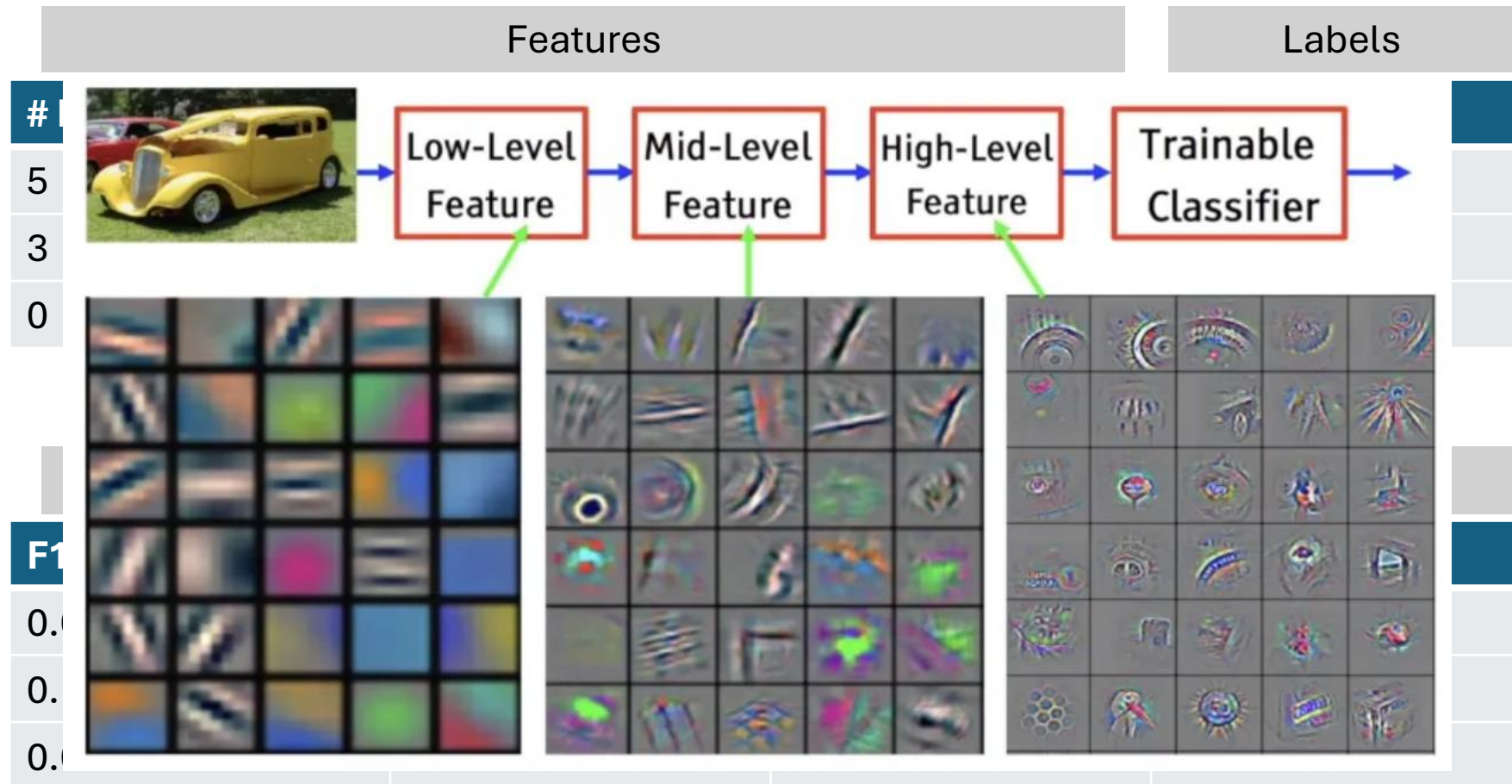


DL Example

Convolution Neural Network (CNN)



Feature Importance



Feature Importance

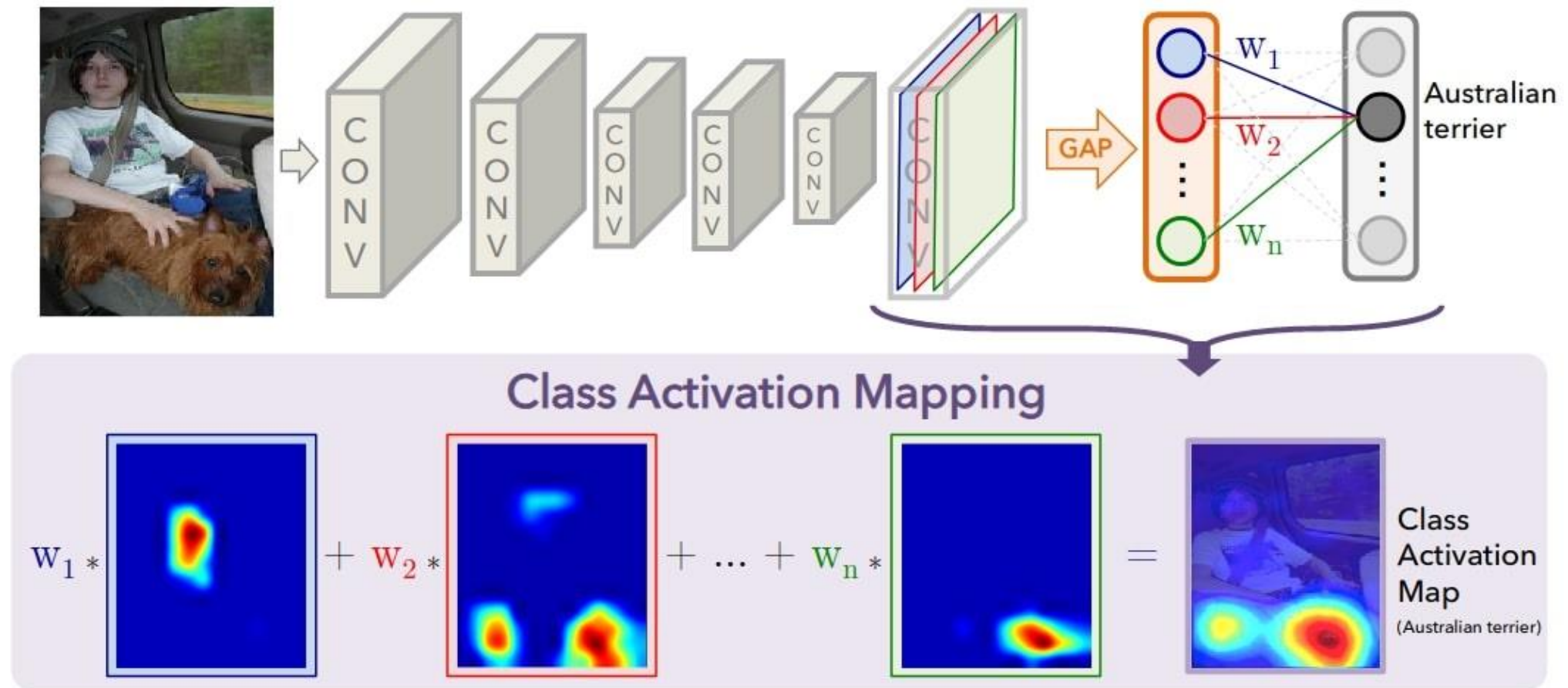


Figure 2. Class Activation Mapping: the predicted class score is mapped back to the previous convolutional layer to generate the class activation maps (CAMs). The CAM highlights the class-specific discriminative regions.

So much data ... but why?

BF1
0
1

BF1	BF2
0	0
0	1
1	0
1	1

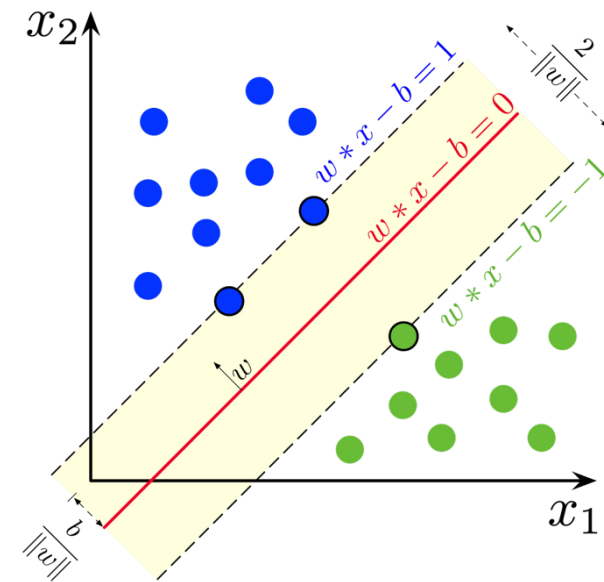
BF1	BF2	BF3
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

BF1	BF2	BF3	BF4
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

Curse of Dimensionality

Id	F1	F2	F3
P1	3	5	10
P2	8	6	4

$$d(P1, P2) = \sqrt{(3 - 8)^2 + (5 - 6)^2 + (10 - 4)^2} = 7.87$$



Id	F1	F2	...	Fn
P1	3	5	...	10
P2	8	6	...	4

$$d(P1, P2) = |3 - 8| + |5 - 6| + |... - ...| + |10 - 4|$$

$$d(P1, P2) = 5 + 1 + \dots + 6 = 6,000,000$$

Id	F1	F2	...	Fn
P1	4	5	...	10
P2	8	6	...	4

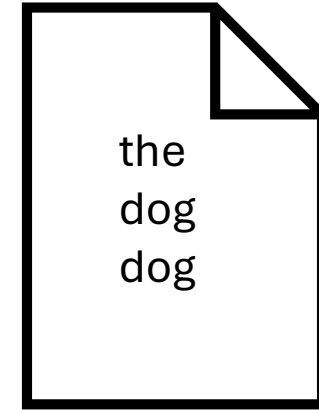
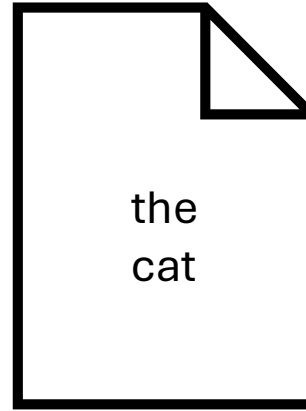
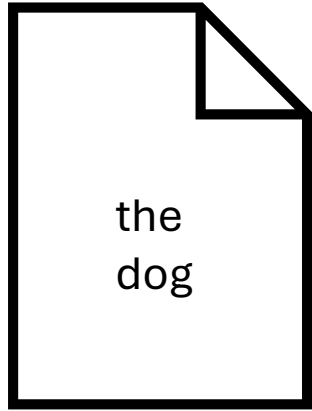
$$d(P1, P2) = |4 - 8| + |5 - 6| + |... - ...| + |10 - 4|$$

$$d(P1, P2) = 4 + 1 + \dots + 6 = 5,999,999$$

!!! In high-dimensional spaces, all points will have the same distance from each other !!!

Large Language Models

Information Retrieval: Finding the best result



Term Frequency (TF)

the = $\frac{1}{2}$
cat = 0
dog = $\frac{1}{2}$

the = $\frac{1}{2}$
cat = $\frac{1}{2}$
dog = 0

the = $\frac{1}{3}$
cat = 0
dog = $\frac{2}{3}$

Inverse Document Frequency (IDF)

the = $\log \frac{3}{3} = 0$
dog = $\log \frac{3}{2} = 0.4$
cat = $\log \frac{3}{1} = 1.1$

Query: "the dog"

tfidf("the", D1) = $\frac{1}{2} * 0 = 0$
tfidf("the", D2) = $\frac{1}{2} * 0 = 0$
tfidf("the", D3) = $\frac{1}{3} * 0 = 0$

+ tfidf("dog", D1) = $\frac{1}{2} * 0.4 = 0.2$
+ tfidf("dog", D2) = $0 * 0.4 = 0$
+ tfidf("dog", D3) = $\frac{2}{3} * 0.4 = 0.26$

Language is hard!

Morphism of words: dog, dogs; go, went

Synonyms

Meaning changes over time

Language is hard!

I shot an elephant in pajamas

Interpretation 1: I wore pajamas when I shot the elephant

Interpretation 2: The elephant wore pajamas when I shot it

➔ Language is vague & ambiguous

Language is hard!

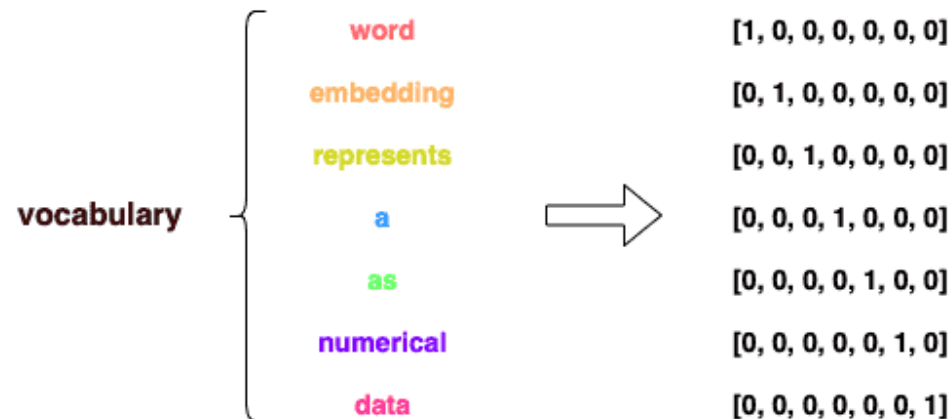
Irony & Sarcasm

“Terrible weather, it’s raining cats and dogs outside!”

Not everything on the internet is true – but what is and what is not?

Understanding Language?: Embedding Model

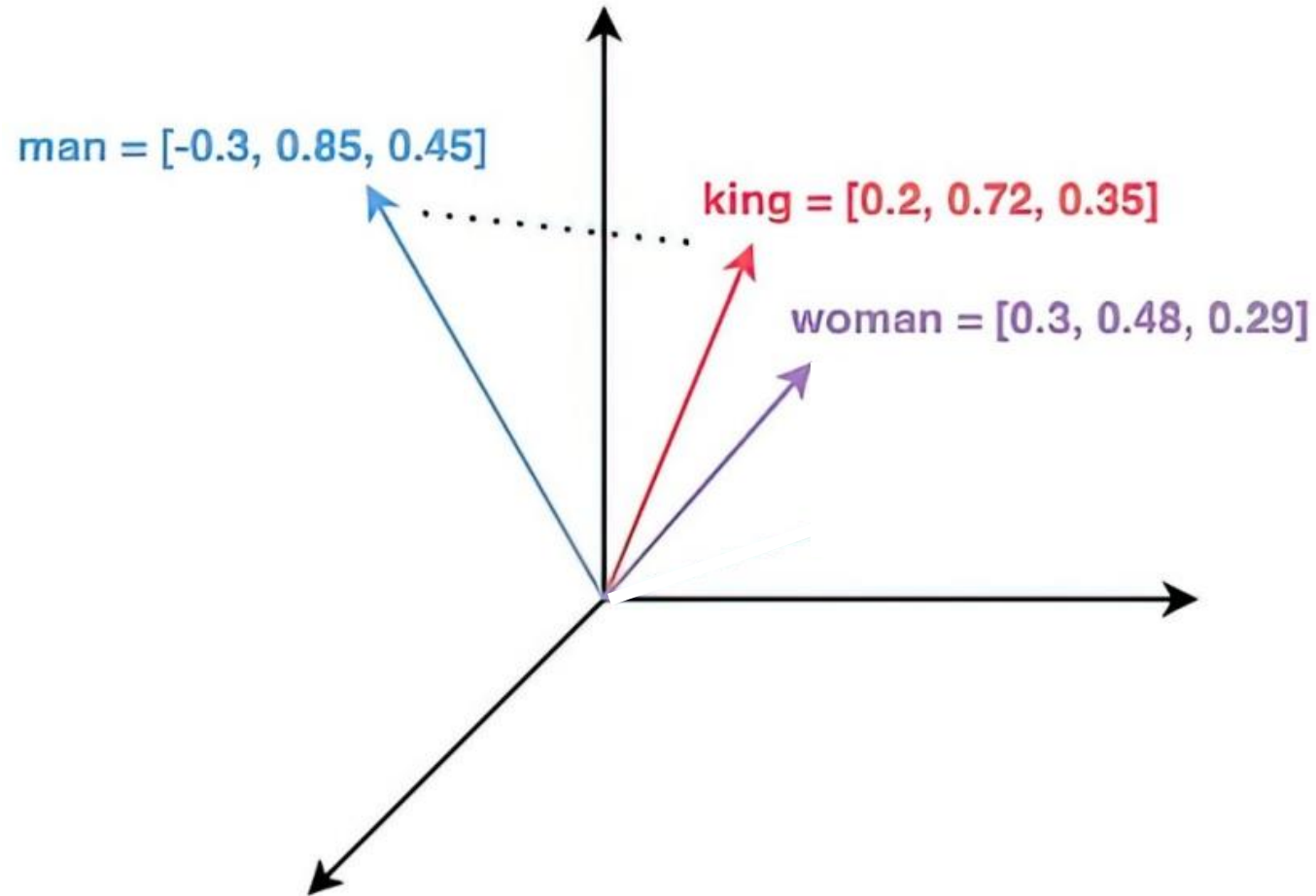
Sentence: Word embedding represents a word as numerical data.



	Document 1	Document 2	Document 3	Document 4	
Word 1	0.85	0	0.3	0	
Word 2	0.34	0.12	0	0.66	Word embedding
Word 3	0	0.73	0.12	0	
Word 4	0.25	0	0.24	0.56	
Word 5	0	0.54	0.2	0.49	

TF-IDF

Understanding Language?: Embedding Model



LLMs: Token Predictors & Temperature

I went to the zoo and saw

lions	(0.2)
hipp	(0.13)
croc	(0.11)
meer	(0.004)

Temperature = 0 → always take highest probability

Temperature > 0 → randomly takes lower probabilities

Temperature 0 does not guarantee the same output!

Floating point errors: Huge dimensionality → very small distances → probabilities can “randomly” shift

Examples: Different GPU types, varying GPU loads

Cascading Effect:

I went to the (zoo|circus|train station) (and) (saw|enjoyed|waited) (croc|elephant|hipp)

Hallucinations: Types

1. Factual Hallucinations

"The University of Arizona was founded in 1881" (actually 1885)

2. Faithfulness Hallucinations

Summarizes a document with the opposite conclusion

3. Temporal Hallucinations

States a regulation as current when it was amended

4. Entity Hallucinations

Fabricates a named executive or a report title

5. Context Hallucinations

Answers about Q3 when Q4 data was retrieved

Agentic AI

Definition

AI agents enhance LLMs and similar generalist AI models by enabling them to automate complex procedures.

They can execute multi-step plans, use external tools, and interact with digital environments to function as powerful components within larger workflows.

Chances & Risks

Automate complex tasks/behavior

Easy to create (using LLMs)

Quick to prototype

Human integration possible via natural language

Non-deterministic behavior (“unreliable”)

Transparency can be difficult

Expensive to run / environmental cost

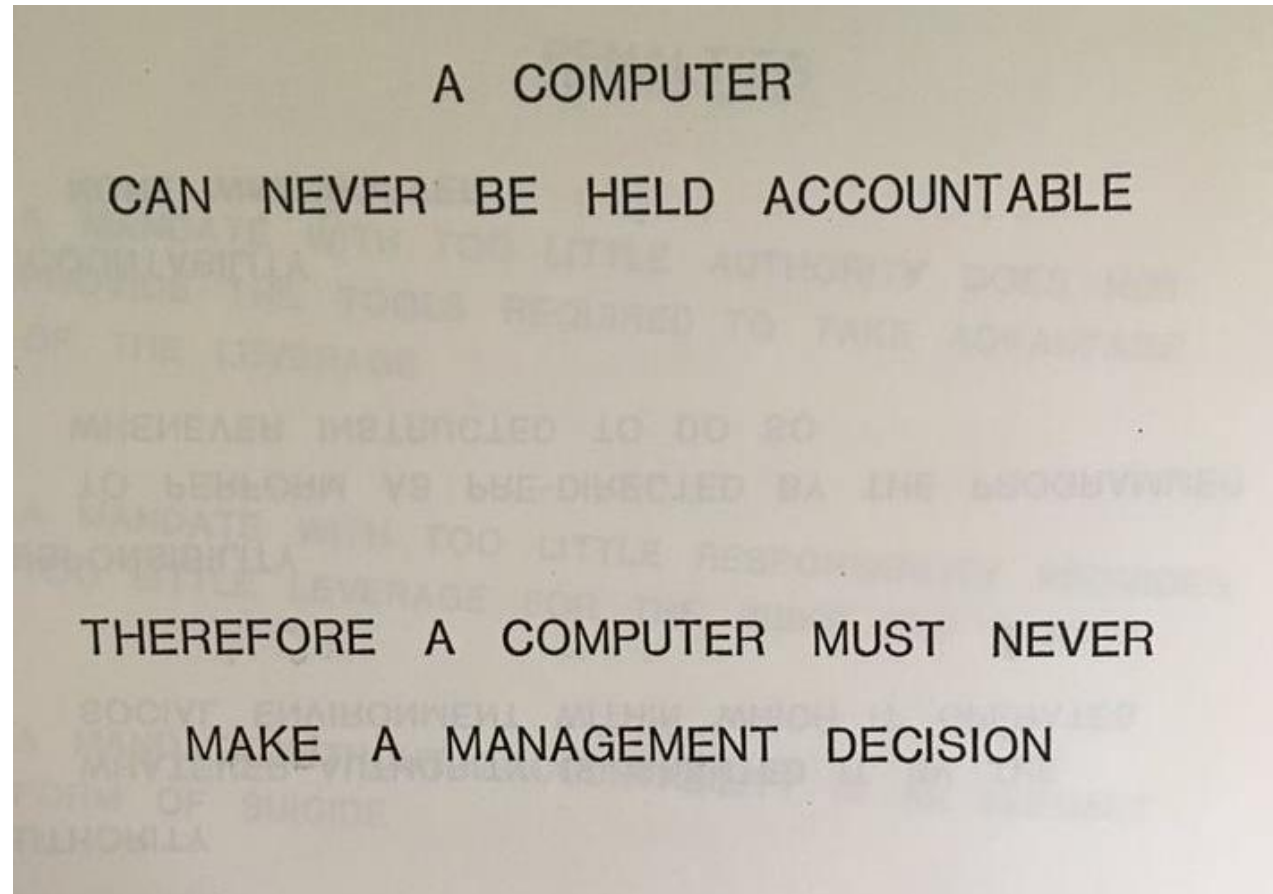
Unethical behavior possible

Cybersecurity and Privacy issues

Unclear Accountability

Accountability

IBM, 1979



Florida sues OpenAI and Sam Altman over alleged safety lapses

JUNE 1, 2026 · 3:52 PM ET

EXCLUSIVE MoneyWatch

Their son died of a drug overdose after consulting ChatGPT. Now they're suing OpenAI.

By [Megan Cerullo](#), [Jo Ling Kent](#), Emily Pandise

Updated on: May 12, 2026 / 12:32 PM EDT / CBS News

Add CBS News on Google



The lawsuit, filed in Florida state court on Monday, accuses OpenAI and its CEO, Sam Altman, of failing to warn users that ChatGPT could be dangerous and instead marketing it as safe and reliable, including for children.

Justin Sullivan/Getty Images

Florida is accusing OpenAI and its CEO Sam Altman of putting profit over safety, in the first lawsuit brought by a state against the ChatGPT maker over the alleged shortcomings of the chatbot.

Source: <https://www.reuters.com/legal/litigation/tesla-settles-wrongful-death-lawsuit-over-crash-that-killed-florida-teenager-2026-04-20/>
<https://www.cbsnews.com/news/chatgpt-lawsuit-colorado-man-suicide-openai-sam-altman/>
<https://www.cbsnews.com/news/open-ai-chatgpt-drug-overdose-lawsuit/>
<https://www.npr.org/2026/06/01/nx-s1-5843132/openai-florida-lawsuit-safety-chatgpt>

Tesla settles wrongful death lawsuit over crash that killed Florida teenager

By [Mike Scarcella](#) and [Abhirup Roy](#)

April 20, 2026 2:41 PM PDT · Updated April 20, 2026



Tesla logo is seen in this illustration taken July 23, 2025. REUTERS/Dado Ruvic/Illustration/File Photo [Purchase Licensing Rights](#)

MoneyWatch

ChatGPT served as "suicide coach" in man's death, lawsuit alleges

By [Mary Cunningham](#)

Updated on: January 15, 2026 / 3:42 PM EST / CBS News

Add CBS News on Google

A new lawsuit filed against OpenAI alleges that its ChatGPT artificial intelligence app encouraged a 40-year-old Colorado man to commit suicide.

IN2DREAMS



Artificial Intelligence Accountability Policy

AI ACCOUNTABILITY CHAIN



Source: NTIA

Bias

THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG
PILE OF LINEAR ALGEBRA, THEN COLLECT
THE ANSWERS ON THE OTHER SIDE.

WHAT IF THE ANSWERS ARE WRONG?

JUST STIR THE PILE UNTIL
THEY START LOOKING RIGHT.



Data Bias

“Data bias means that, regardless of the particular model used, any statistical prediction model relying on pattern recognition will exhibit the same kind of biases present in the training data.”

Amazing new AI Model to predict Rain in Tucson: 95.9 % correct!

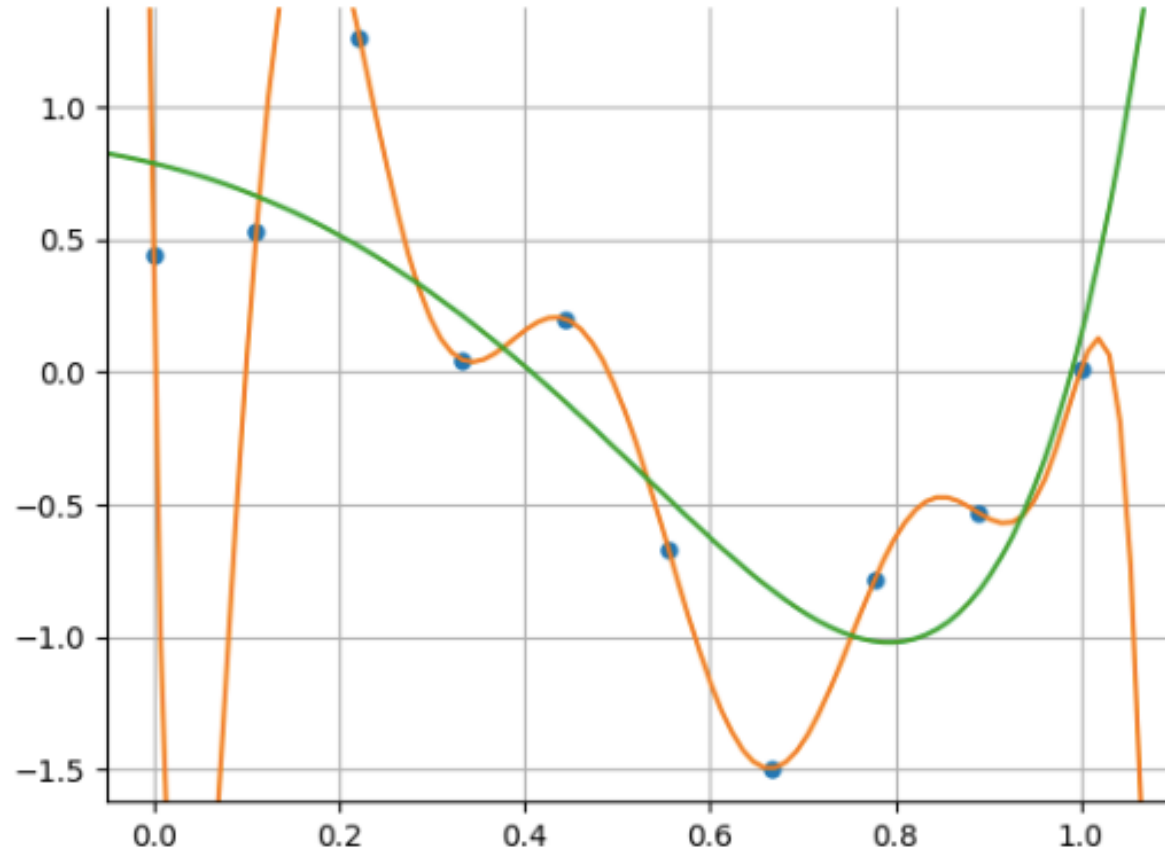
```
1
2
3  def will_it_rain_in_tucson():
4      |   return "no"
5
```

The definition of insanity is doing the same thing over and over and expecting different results.

- Albert Einstein

Assuming 15 days of rain per year over the last 10 years: $350 / 365 = 95.9 \%$

Model Bias



Human-In-The-Loop-Bias

- Overtrust in AI system
 - Uber safety driver distracted → deadly crash → could have been prevented
- Spot-checking citations
 - 5 look good → 3 others were hallucinated
- Confirmation bias
 - AI outcomes aligning with own expectations convince us more easily

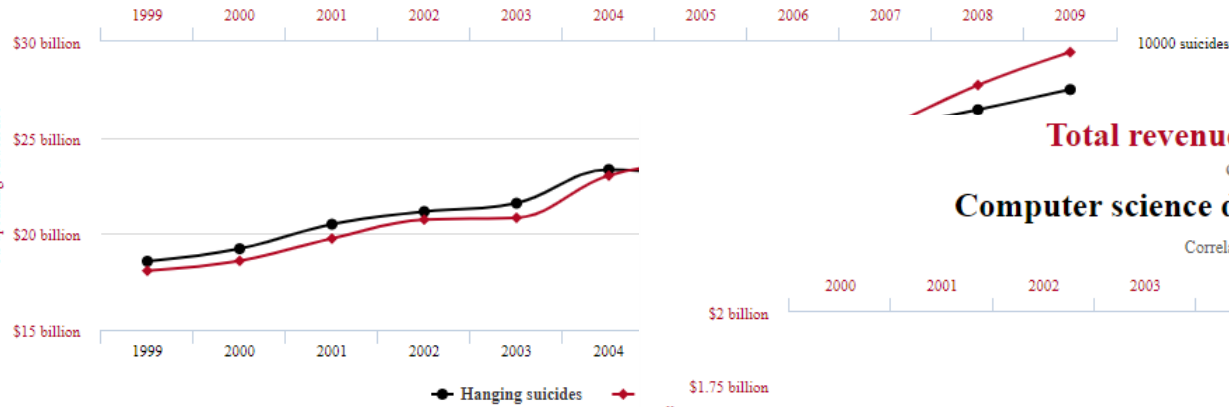
Spurious Correlations

US spending on science, space, and technology

correlates with

Suicides by hanging, strangulation and suffocation

Correlation: 99.79% ($r=0.99789126$)



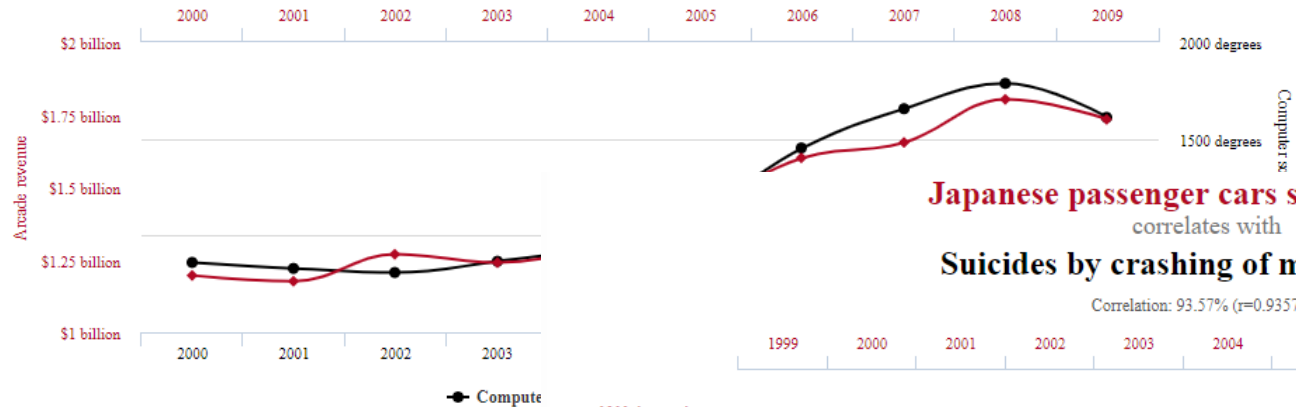
Data sources: U.S. Office of Management and Budget and Centers for Disease Control & Prevention

Total revenue generated by arcades

correlates with

Computer science doctorates awarded in the US

Correlation: 98.51% ($r=0.985065$)



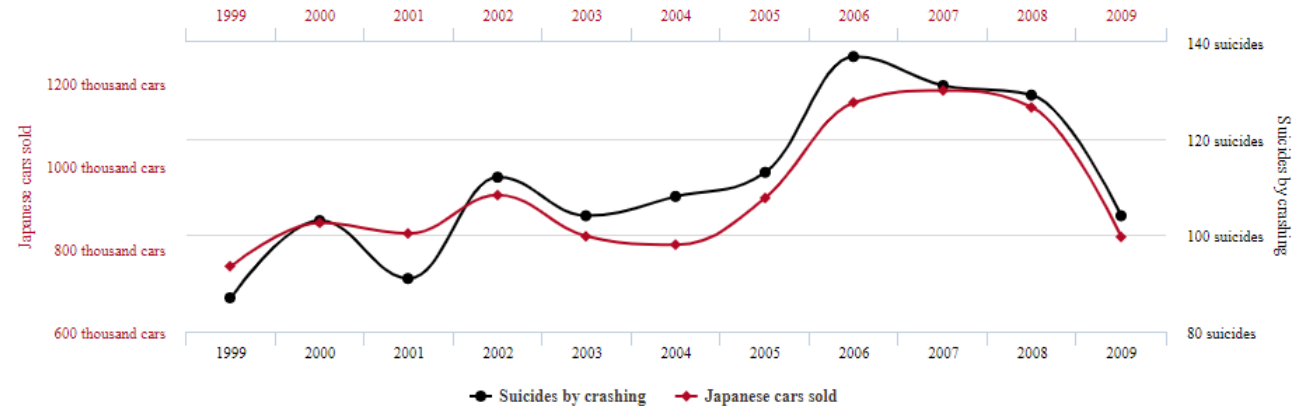
Data sources: U.S. Census Bureau and National Science Foundation

Japanese passenger cars sold in the US

correlates with

Suicides by crashing of motor vehicle

Correlation: 93.57% ($r=0.935701$)



Data sources: U.S. Bureau of Transportation Statistics and Centers for Disease Control & Prevention

Correlation vs. Causality

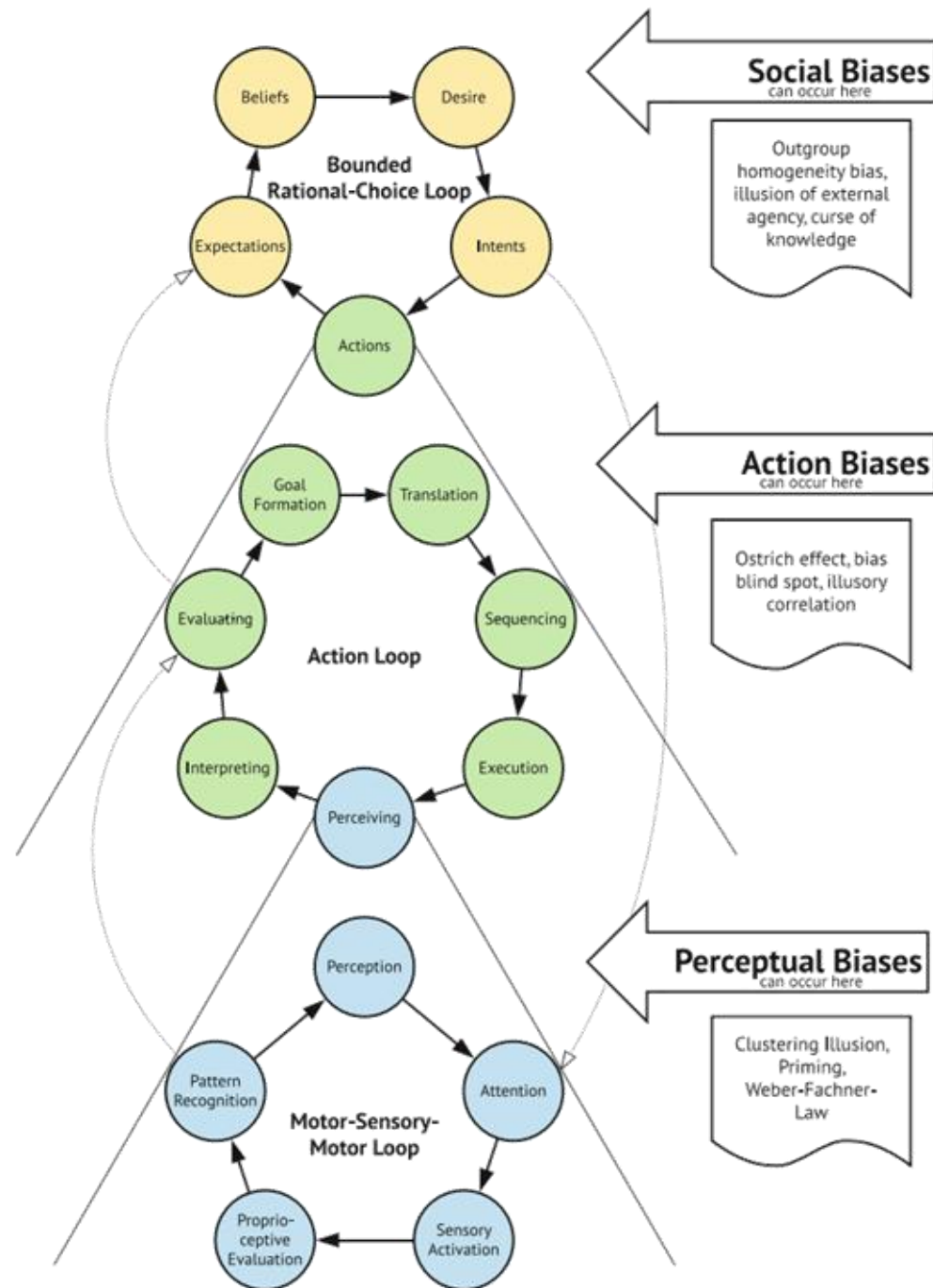
Every time I eat ice cream, I get sunburned

Wrong: Ice cream causes sunburn

Correct: Hidden causality: sunny/hot

→ eat ice cream

→ get sunburned



Geoffrey Ellis *Editor*

Cognitive Biases in Visualizations

Biases Mitigation Strategies

Awareness & Recognition

Aligning data bias (not overly favor one class/outcome)

Alternative performance metrics

Risks

Ethical and Economical Implications

People lose jobs*

Degrading quality or unreliable products/services

AI is more biased than assumed/acknowledged

Deepfakes accelerate distrust in societies

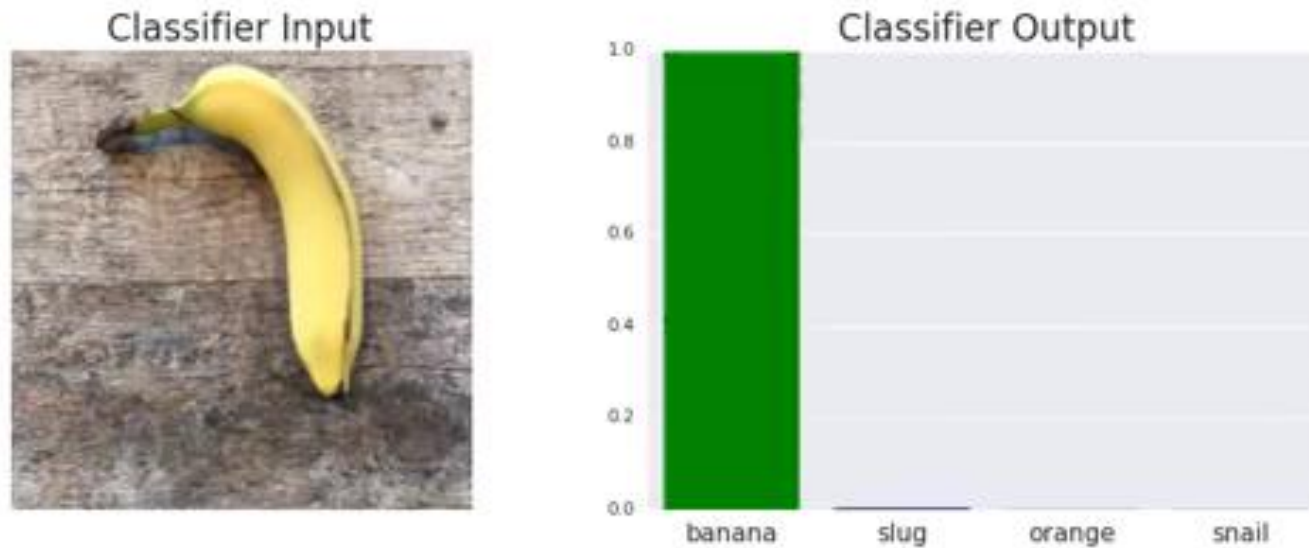
Privacy & Security

AI and automation require/favor data access

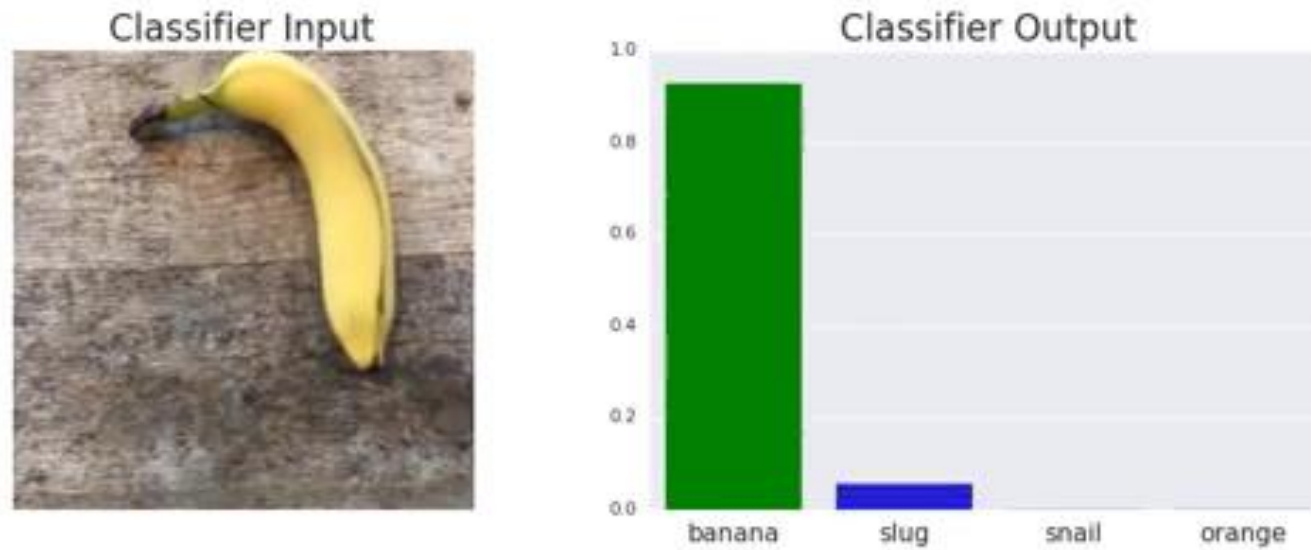
→ Combining multiple data sources allows de-anonymization, tracking

→ AI can create new attack vectors into

Adversarial attacks



Adversarial attacks



Adversarial attacks



Granny Smith	85.6%
iPod	0.4%
library	0.0%
pizza	0.0%
toaster	0.0%
dough	0.1%

Sustainability

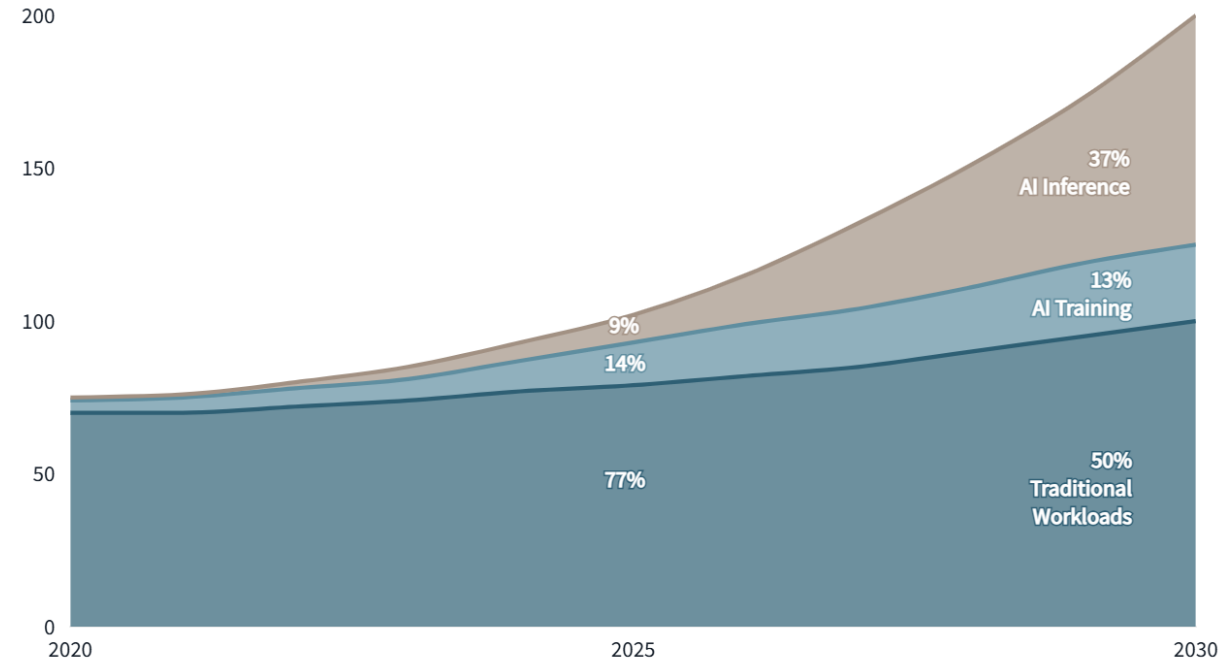
Current frontier models require massive amounts of computational power

We are currently in the \$4 Uber-era

A significant shift is anticipated in 2027, when inference workloads could overtake training as the dominate AI requirement

Total global data center workloads (GW)

■ Traditional workloads ■ AI Training ■ AI Inference



Source: JLL Research, 2025

Copyright Law

OpenAI loses bid to dismiss part of US authors' copyright lawsuit

By Blake Brittain

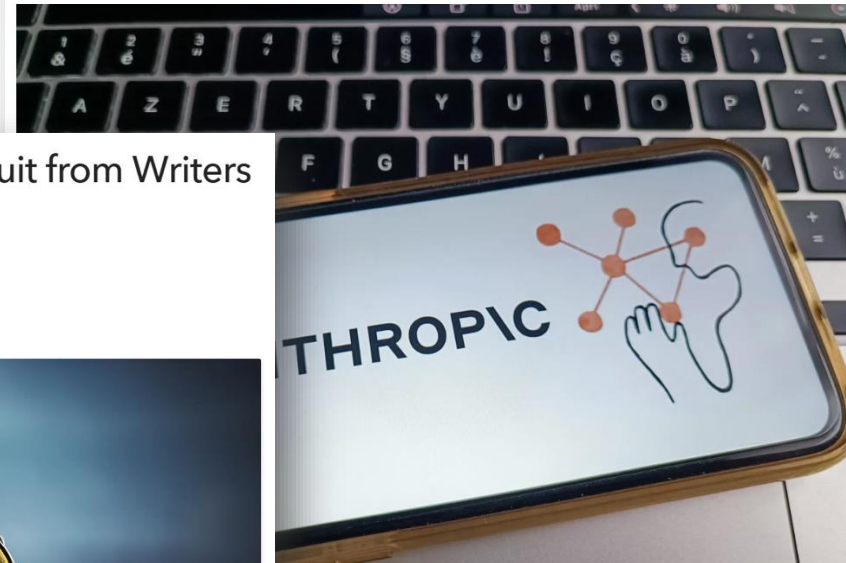
October 28, 2025 1:35 PM PDT · Updated October 28, 2025



Anthropic settles with authors in first-of-its-kind AI copyright infringement lawsuit

SEPTEMBER 5, 2025 · 8:19 PM ET

 Chloe Veltman



nt by a group of authors was settled on Friday.
y Images

 OpenAI

ChatGPT

OpenAI and ChatGPT logos are seen in this illustration taken, February 3, 2023. REUTERS/Dado Ruvic/Illustration [Purchase Licensing Rights](#)

OpenAI, Anthropic, xAI Hit With Copyright Suit from Writers (2)

December 22, 2025, 3:16 PM PST; Updated: December 23, 2025, 9:48 AM PST

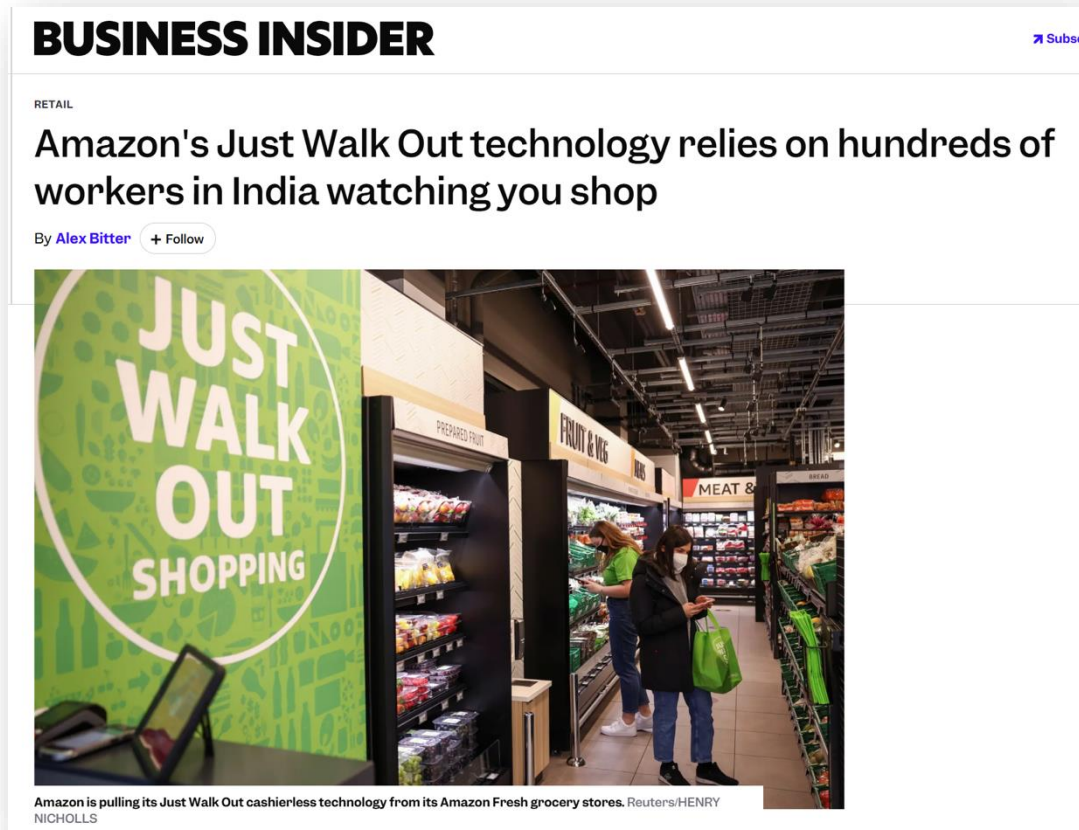
 **Annelise Levy**
Reporter



Photo Illustration: Braden Brook/Bloomberg Law; Photos: Getty Images.

Source: <https://news.bloomberglaw.com/ip-law/openai-anthropic-xai-hit-with-copyright-lawsuit-from-writers>
<https://www.reuters.com/legal/government/openai-loses-bid-dismiss-part-us-authors-copyright-lawsuit-2025-10-28/>
<https://www.npr.org/2025/09/05/nx-s1-5529404/anthropic-settlement-authors-copyright-ai>

Deceiving AI / Human exploitation



Anthropomorphizing computers

Hallucinations, Thinking, Reasoning

→ Attaching human-like qualities to AI & robots

→ Can lead to hype & overtrust

Responsible AI usage

Policy

Be aware of various policies if/how AI usage is allowed

- Institution

- Funding Agency

- Journals & conferences

- Instructors: Clearly define AI usage in syllabi

Impact

Be aware of the impact of using AI

What if it goes right?

What can I achieve?

What if it makes mistakes?

Can I / someone catch them?

Do they matter?

Is it worth it?

Costs

Ethical reasons

Verification & Bias check

Verify all AI-generated output or modifications for correctness

Check for potential biases

- Are certain outcomes preferred?

- Alternative perspectives missing?

- Embedded assumptions in the data/model?

➔ High chance that you will be held accountable!

Transparency

Always declare if and how AI has been used.

Disclose:

Tools

Versions

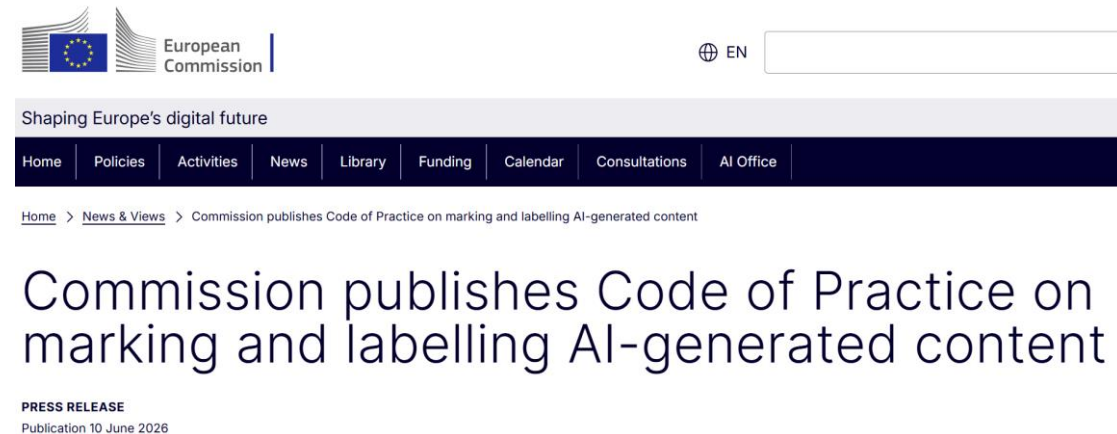
Purposes

Document (for yourself / for audits):

AI Engagements

Source-code

Chat histories + AI-generated material



Privacy

Protect sensitive information such as PII, and anonymize data as much as possible

Understand retention policies for various data types

Follow institutional guidelines and use institutionally approved tools, preferably

Human Agency

AI augments – never replaces human judgment

Remain the ultimate decision maker

Maintain critical thinking and domain expertise



Responsible AI: Transparency, Accountability, Bias

Thank you!